I. INDEPENDENT GAUSSIAN NOISE AS A CONTROL PERTURBATION

A. Rationale

Psilocybin has been associated with increased neural signal diversity and has motivated accounts of psychedelic states in terms of increased entropy or variability [1, 2]. These findings motivate testing whether a working-memory network is sensitive to stochastic perturbation, but they do not imply that psilocybin acts as additive Gaussian noise. We therefore used independent Gaussian noise as a deliberately non-specific control rather than as a mechanistic model of the drug. This choice also has a direct computational rationale: stochastic fluctuations can drive diffusion of maintained representations, and the extent of this drift depends on the stability and geometry of the recurrent memory dynamics [3, 4]. The control consequently asks a restricted question: how much degradation follows from adding unstructured variability of a known magnitude to an otherwise unchanged trained network?

B. Implementation

Noise was added to the hidden-state update only during the memory delay. For each active time step t, trial b, and hidden unit i, an independent sample $\eta_{tbi} \sim \mathcal{N}(0,1)$ was drawn. The realised perturbation was

$$\varepsilon_{tbi} = \sigma \frac{\eta_{tbi}}{\left(|\mathcal{A}|^{-1} \sum_{(t,b,i)\in\mathcal{A}} \eta_{tbi}^2\right)^{1/2}}, \tag{1}$$

where $\mathcal{A}$ denotes all delay-active entries and σ is the target root-mean-square (RMS) perturbation strength. This normalisation matched the realised RMS exactly within each evaluated batch. Noise was independent across time, trials, and units; no temporal autocorrelation or population covariance was imposed. Five independently trained frozen models were evaluated at delay lengths of 20, 80, and 160 steps and at RMS strengths 0, 0.01, 0.025, 0.05, and 0.10. Each model–delay condition comprised five stochastic replicates of 20 paired 64-trial batches. A linear decoder fitted to 512 clean trials per model and delay quantified maintenance error, and the trained model seed was treated as the replication unit ($n = 5$). RMS is an intervention parameter and should not be interpreted as a pharmacological dose.

C. Results

Gaussian noise produced a graded, delay-dependent impairment (Fig. 1). At RMS 0.05, mean response error increased from $4.34 \pm 1.54°$ to $4.91 \pm 1.45°$ at delay 20, from $6.49 \pm 1.93°$ to $7.47 \pm 1.68°$ at delay 80, and from $9.03 \pm 2.50°$ to $10.44 \pm 2.24°$ at delay 160 (mean ± SD across five model seeds). The impairment became clearer at RMS 0.10, where the corresponding response errors were $6.42 \pm 1.35°$, $10.48 \pm 1.92°$, and $14.68 \pm 2.57°$.

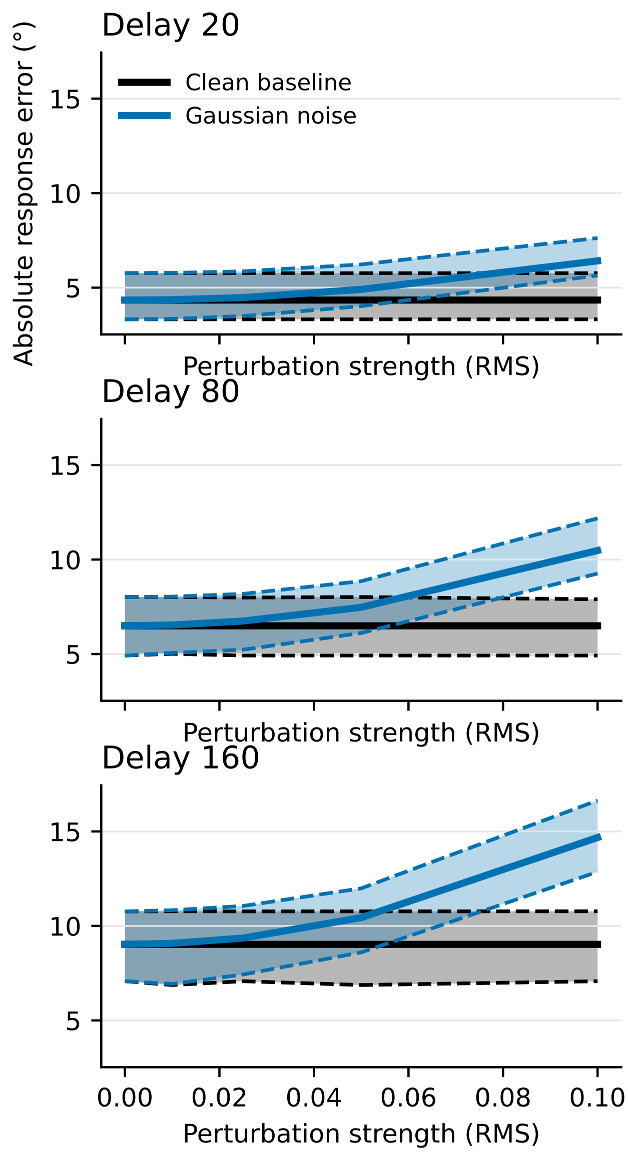

FIG. 1. **Behavioural response to independent Gaussian perturbation.** Absolute response error is shown across perturbation strengths and delay lengths. Solid lines are five-model means; dashed lines and shading denote 95% bootstrap confidence intervals across models.

The maintenance measures showed a larger relative effect than the final behavioural readout (Fig. 2). At delay 80 and RMS 0.05, clean-decoder error increased from $1.29 \pm 0.82°$ to $3.71 \pm 1.53°$, while decoded angular drift increased from $2.60 \pm 1.84°$ to $5.20 \pm 2.11°$. Fixation/wait–go accuracy was effectively preserved (0.98 ± 0.00 in both conditions), indicating that the perturbation primarily affected the maintained angle rather than task-phase gating. The decoder was fitted to clean hidden states and then evaluated on the corresponding perturbed states, so clean-decoder error measures representational distortion rather than a refitted noisy readout.

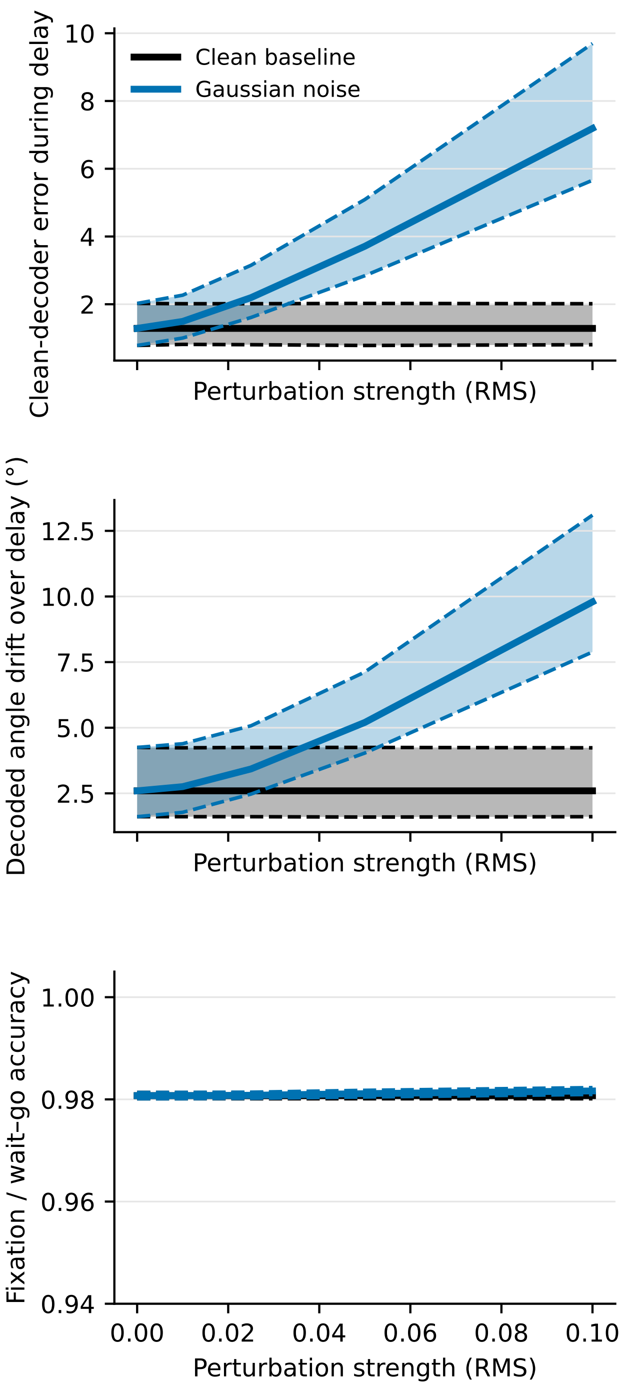

FIG. 2. **Maintenance effects at delay 80.** Clean-decoder error and decoded drift increased with Gaussian-noise RMS, whereas fixation/wait–go accuracy remained stable. Lines and intervals are defined as in Fig. 1.

The geometry analysis was consistent with mild blurring rather than collapse of the circular representation (Fig. 3). At delay 80 and RMS 0.05, within-angle dispersion increased from 0.550 to 0.732, whereas normalised between-angle centroid spacing was 0.990 relative to the clean value of 1. Thus, trial-to-trial states for the same remembered angle became more dispersed while the global separation of different angles was largely retained.

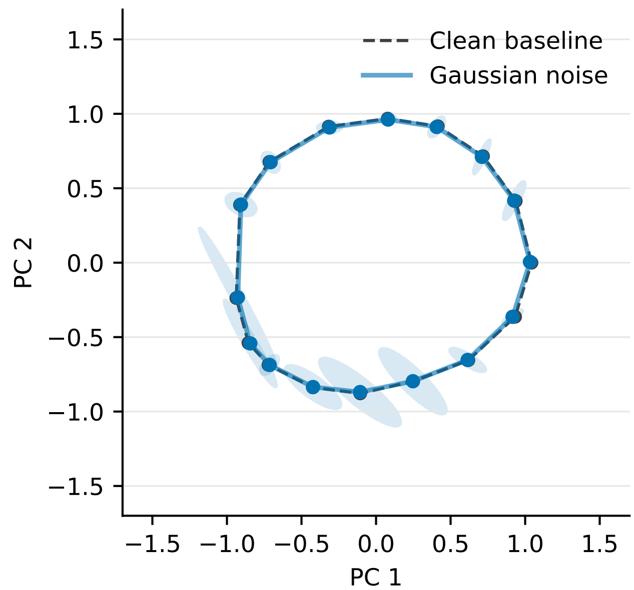

FIG. 3. **Overlay of clean and Gaussian memory geometry at delay 80 and RMS 0.05.** Angle centroids and uncertainty ellipses are shown in the clean-fitted PC1–PC2 plane. The Gaussian ring is overlaid on the clean baseline so that its mild within-angle broadening and small centroid changes can be read directly.

Response settling was close to its measurement floor at the primary RMS (Fig. 4). At delay 80, the mean first response step within 20° of the target increased only from 5.02 in the clean condition to 5.19 at RMS 0.05; 99.2% of trials settled. At RMS 0.10, settling increased to 6.55 steps and the settled fraction fell to 93.6%. Final response error in this focused analysis increased from 5.94° clean to 6.75° at RMS 0.05 and 10.40° at RMS 0.10.

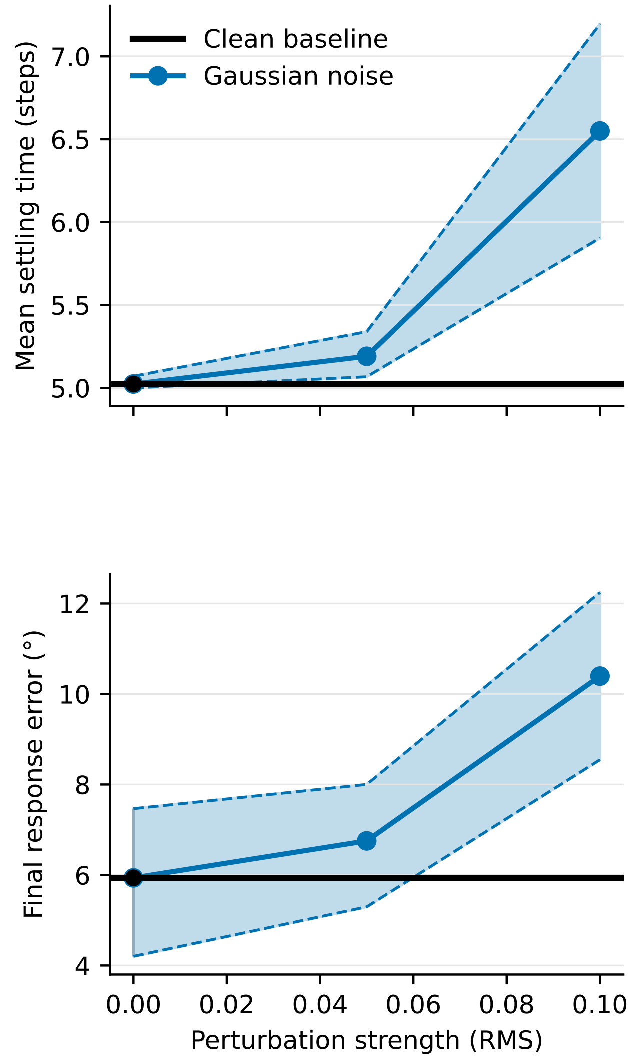

FIG. 4. **Response settling after the response cue at delay 80.** Settling was defined as the first counted response step with absolute circular error below 20°; the first five transition steps were excluded. Points denote trained models and lines denote five-model means.

Together, these results show that independent Gaussian noise is useful as a weak, non-specific degradation control. Its effects scale with perturbation strength and maintenance duration, and are expressed most clearly as increased decoder error, drift, and within-angle dispersion. The result establishes a reference against which more targeted perturbations can be judged; it does not provide evidence that Gaussian noise itself recreates psilocybin's neural or behavioural effects.

[1] R. L. Carhart-Harris *et al.*, "The entropic brain: a theory of conscious states informed by neuroimaging research with psychedelic drugs," Front. Hum. Neurosci. **8**, 20 (2014), doi:10.3389/fnhum.2014.00020.

[2] M. M. Schartner, R. L. Carhart-Harris, A. B. Barrett, A. K. Seth, and S. D. Muthukumaraswamy, "Increased

spontaneous MEG signal diversity for psychoactive doses of ketamine, LSD and psilocybin," Sci. Rep. **7**, 46421 (2017), doi:10.1038/srep46421.

[3] Z. P. Kilpatrick, B. Ermentrout, and B. Doiron, "Optimizing working memory with heterogeneity of recurrent cortical excitation," J. Neurosci. **33**, 18999–19011 (2013), doi:10.1523/JNEUROSCI.1641-13.2013.

[4] E. Ghazizadeh and S. Ching, "Slow manifolds within network dynamics encode working memory efficiently and robustly," PLoS Comput. Biol. **17**, e1009366 (2021), doi:10.1371/journal.pcbi.1009366.